

Praveen
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AI AGENT ENGINEER & SYSTEMS ARCHITECT

CONTACT

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AGENT SYSTEMS

Orchestration
LangGraph workflows · DSPy signatures · tool calling and memory · FastMCP servers · Postgres and Celery job queues

Streaming & HIL
WebSocket and SSE state streaming · approval gates

Production AI
prompt-injection guardrails · model gateway, retries, fallback · token budgets · Gemini, Claude, Azure OpenAI

Evaluation
LLM-as-judge (MQM) · offline evals: chrF, cost, latency · red-team runs

Coding agents
Claude Code · autonomous issue-to-release agent · AI code review

STACK

Python · TypeScript · FastAPI · Next.js · PostgreSQL / pgvector · SQLAlchemy · Celery / Redis · FastMCP · AWS ECS · Terraform · Docker

SELECTED IMPACT

1,300+ platform users, sole architect
30 MCP tools on one service layer
25 exec sessions run over MCP
3 eng pods on AI code review

LANGUAGES

English (professional) · Mandarin (learning)

PROFILE

AI engineer who ships agent systems end to end: LangGraph and DSPy orchestration, MCP servers, durable job queues, model gateways, state streamed to the UI behind approval gates, and the guardrails and evals that keep them safe. Six years of production AI in Python and TypeScript; now forward-deployed in ViewSonic's CEO Office, several workstreams at once, with US and UK counterparts.

EXPERIENCE

Forward Deployment Pod Lead (hands-on)
Jul 2026 – Present

ViewSonic • CEO Office | Taipei, Taiwan

- **GMA transformation.** Lead the Global Marketing Agency's new tooling and build its core: a service cost calculator (any brief in, DSPy extraction to priced lines against a Finance-governed rate card, approval freezes prices, recovery tracked against target) and a translation engine; FastAPI, Next.js 16, Postgres/pgvector, CI to AWS ECS with migration and smoke-test gates.
- **Durable jobs, model gateway.** Translation jobs on a Postgres queue: leased and renewed, writes fenced on the attempt, three attempts with backoff, crash recovery redrafts only in-flight batches. The gateway retries 429/5xx, falls back to a backup model, halves a batch at the output ceiling, holds per-job token budgets.
- **Prompt-injection guardrails, HIL.** Brief text is untrusted: a canary in every system prompt, every model claim re-resolved against the catalogue (an injected id becomes an unmapped line, never an action), the model never sets a price. Account managers verify each line before approval; corrections teach a mapping memory.
- **Evaluation and live state.** A second model scores every draft with MQM issues, never the one that wrote it; offline chrF, flag and refusal rates, cost per model and locale, red-team of hostile copy. WebSocket pushes a full snapshot after every mutation; thirty brief-shielded FastMCP tools share the REST service layer.
- **Product sense.** Asset Management Tool for GMA (batch-renames creative assets for Bynder): PRD, living decision register and UX prototype on the ViewSonic design system; a pure rules engine with position-aware parsing, tests derived from the convention, pluggable source and destination adapters.
- **Autonomous coding agent** on the calculator: picks up an issue, writes and tests the fix, clears every gate, releases to dev unaided; human approval gates production. AI code review across three pods; Claude skills for decks and meetings.
- **Platform MCP server** (SSO-bound, role-scoped tools) so staff query goals, AI-maturity and org data from Claude; an MCP workflow on it runs a 25-session leadership offsite in London this month.

Senior AI Systems Architect
Mar 2022 – Jun 2026

ViewSonic • CEO Office | Taipei, Taiwan

- **Sole architect of a 1,300+ user AI platform**, blank repo to production: LangGraph stateful workflows with per-user memory, tool calling, Text-to-SQL, pgvector RAG, Celery/Redis long-running jobs, SSE streaming to a Next.js UI.
- **Guardrails and evals.** Full OWASP LLM Top 10 coverage, per-user answer redaction at inference, AST validation on generated SQL, versioned ethics gate; OKR criteria as LLM-validated rubrics, an L1 to L5 competency model with behavioural anchors. Rule: *prompts for judgment, code for determinism.*
- **Gemini Vision pipeline** (PPTX, DOCX, PDF, images to brand-compliant decks) with an AI auditor that scores six dimensions and triggers remakes; distilled a 600M model to 2× faster at parity; 8-GPU Kubernetes cluster for PyTorch.

AI Software Developer
Aug 2020 – Mar 2022

Neurelli | Taipei, Taiwan

- Shipped industrial computer vision to production (licence-plate recognition on TensorRT/CUDA, LCD defect detection), from data to deployment.

EDUCATION

M.Sc. Electrical Engineering & Computer Science
2018 – 2020

National Taipei University of Technology | Taiwan

- Thesis: CUDA-accelerated flood simulation, 15× speedup; published at IEEE.